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Seat No. _____

No. of Printed Pages: 2



SARDAR PATEL UNIVERSITY

M. Sc. (IT) INTEGRATED NC EXAMINATION, VI SEM

Date : 09th Apr 2022, Saturday

Time : 12:00pm To 02:00pm

Session : NOON

Sub: Artificial Intelligence

Course No : PS08CIIT03

Total marks : 70

Q – 1 (a) Select an appropriate option

[08]

- i) _____ is a software program that infers the knowledge available in the knowledge base.
 - a) Knowledge base
 - b) Inference engine
 - c) User Interface
 - d) None of above
- ii) _____ represent sets of raw observation, alphabets, symbols or statements.
 - a) Facts
 - b) Rules
 - c) Heuristics
 - d) None
- iii) In _____ AI language, facts and procedures are developed and presented in the form of list.
 - a) Java
 - b) PROLOG
 - c) LISP
 - d) None
- iv) The meaning of conjunction is given by _____ function.
 - a) Union
 - b) Intersection
 - c) Negation
 - d) both b & c
- v) A _____ neural network in which there is at least one feedback loop from output layer to input layer.
 - a) Feed backward
 - b) Feed Forward
 - c) Recurrent
 - d) Nne
- vi) In fuzzy connectives implications are obtained by _____.
 - a) Negation & Intersection
 - b) Intersection
 - c) Union
 - d) Negation
- vii) _____ is a collection of such genes representing possible solutions in domain space.
 - a) Genetic Cycle
 - b) Genetic algorithm
 - c) Mutation
 - d) Genotype
- viii) Genes that are not strong are combined with their mates and used to generate new individuals in a process known as _____.
 - a) Recombination
 - b) crossover
 - c) Mutation
 - d) genotype

Q – 1 (b) Fill in the Blanks and True – False

[16]

- i) Medical diagnosis is a _____ Task.
- ii) Making a machine intelligent is known as _____.
- iii) Heuristics means Algorithm. (True / False)
- iv) Common sense reasoning is an Expert type of task (True / False)
- v) _____ process emphasizes capturing experts thought and experiences.
- vi) Factual knowledge uses _____ symbol for 'implies'.
- vii) A Script organizes knowledge according to cause-and-effect relationships. (True / False)
- viii) Self learning is known as automatic learning or Machine learning. (True / False)
- ix) The meaning of disjunction is given by an _____ function.
- x) _____ converts the fuzzy value into a crisp value.
- xi) The learning rule of perceptron is also known as Delta rule. (True / False)
- xii) A perceptron may contain more than one layer to solve complex real world application. (True / False)
- xiii) A _____ network is a model representation based on reasoning with uncertainly, which is statistical approach.

- xiv) The degree of _____ is referred to the strength of the genes in an individual.
- xv) Inversion is a genetic operation which does not change the solution but change chromosomes itself. (True / False)
- xvi) DNA consist of string built by genes responsible for some property of human to which it belongs. (True / False)

Q – 2 Short Answer attempt **any seven** (Each carry 2 marks) **[14]**

- i) Define AI and write it's characteristics.
- ii) Draw diagram of Data Pyramid according to management perspective.
- iii) Explain the difficulties in KBS development.
- iv) Draw a diagram of development methodology for a Knowledge Based System.
- v) Define Fuzzification and Defuzzification.
- vi) What is Artificial Neural Networks.
- vii) Define chromosomes and Genotype
- viii) What are steps of genetic cycles?
- ix) What is stopping criteria in function optimization?

Q – 3 Write note on the measures for testing the intelligence and level of machine understanding. **[08]**

OR

Q – 3 Explain Applications of Knowledge Based System. **[08]**

Q – 4 Explain existing techniques for Knowledge acquisition. **[08]**

OR

Q – 4 Explain factual knowledge in detail. **[08]**

Q – 5 Explain Fuzzy propositions and inference. **[08]**

OR

Q – 5 Explain the neural network architecture. **[08]**

Q – 6 Write introduction and basic operators of a Genetic Algorithm **[08]**

OR

Q – 6 Minimize a simple function $f(x)=x*x$ on the integer interval $[0,30]$ using genetic algorithm. Take the 01101, 11000, 01000, 10011 strings as individuals in the first population **[08]**

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